

Bachelor's Thesis

Network-Weighted Smoothing of Functional Data

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Abstract

This thesis investigates the estimation of node-specific functional observations when the observation units are connected by a known graph. The proposed network-weighted smoother combines temporal P-spline regularisation with a graph-Laplacian penalty that encourages neighbouring nodes to have similar functional deviations. The method is implemented in R within a functional additive mixed-model framework and is compared with raw observations, complete pooling, and independent nodewise smoothing.

Its performance is assessed in a reproducible Monte Carlo study on a 5×5 lattice with $N = 25$ nodes. The 16-cell core design varies the structure and strength of node-specific signals, the supplied neighbourhood graph, and the observation-noise level. Across 200 replications per cell, the complete network-weighted estimator has lower aggregate mean integrated squared error than separate nodewise smoothing in every core setting. Relative reductions range from 1.4% to 20.5%, with larger gains generally occurring at the higher noise level and for coordinate-smooth signals. A cluster-boundary diagnostic shows that this gain can be attenuated at genuine discontinuities. An $\alpha = 0$ diagnostic is consistent with some of the improvement reflecting general joint shrinkage in addition to graph-specific borrowing. However, a complementary fixed rewired-graph control shows that replacing correctly aligned edges by a degree-preserving rewired graph worsens reconstruction in two selected settings. Taken together, these diagnostics are consistent with general joint shrinkage contributing to the network estimator's advantage and, in the selected rewired-graph settings, with an additional benefit from alignment between the supplied graph and the functional signal.

The method is additionally illustrated using daily mean temperatures from 25 German Weather Service stations and weekly COVID-19 incidence in Bavarian counties. Because the latent functions are unavailable in both applications, the results are interpreted descriptively rather than as external accuracy validations.

Overall, the results support the complete joint approach as a useful reconstruction method when node-specific curves are expected to share structure and graph-informed regularisation is plausible, while emphasising cautious interpretation of graph-specific effects.

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1 Introduction

Functional data arise when each observational unit is represented by a curve rather than by a single scalar value (Ramsay and Silverman, 2005). Examples include incidence trajectories for administrative regions, pollution profiles at monitoring stations, and measurements recorded over time by connected sensors. Such curves are observed only at finitely many points and are often contaminated by noise. Their estimation therefore requires regularisation over the functional argument, which in this thesis is time.

In many applications, however, temporal smoothness is not the only available source of information. The observational units may be connected through a known spatial or administrative neighbourhood structure. Curves at adjacent units may then be more similar than curves at unrelated units, as is commonly considered in spatial functional data analysis (Delicado et al., 2010; Zhang et al., 2016). Smoothing each curve independently ignores this information, whereas pooling all observations into a single curve removes potentially important node-specific variation. Network-weighted smoothing provides an intermediate approach: it retains a separate curve for each node while allowing related nodes to borrow information from one another.

This thesis studies such an estimator for one noisy functional observation at each node of a known graph. The estimator combines a conventional temporal smoothness penalty with an unnormalised graph-Laplacian penalty that penalises differences between the curves of adjacent nodes. It is formulated as a tensor-product smooth within the functional additive mixed-model framework (Scheipl et al., 2015; Greven and Scheipl, 2017) and implemented in R using `refund::pffr()` and `mgcv`. The target is reconstruction of the curves at the observed nodes; prediction at previously unobserved nodes is not considered.

The central research question is whether the complete joint estimator, which combines a functional mean, jointly selected temporal smoothing parameters, and graph-penalised node-specific deviations, improves reconstruction of node-specific functional signals relative to independent nodewise smoothing. A controlled simulation study examines how this comparison varies with the structure and strength of the node-specific signal, the observation-noise level, and the neighbourhood graph supplied to the estimator. The study evaluates the performance of the complete joint estimator. A separate $\alpha = 0$ zero-structured-signal diagnostic assesses the extent to which an advantage over nodewise smoothing can arise from general joint shrinkage. A complementary fixed rewired-graph control then examines, in selected settings, whether replacing the supplied graph by a degree-preserving graph with rewired edges degrades reconstruction.

The thesis makes three contributions. First, it formulates a graph-penalised functional estimator that combines a smooth functional mean with centred node-specific deviations. Second, it provides an implementation that converts graph information into the required neighbourhood representation and aligns curve identifiers with graph nodes before model fitting. Third, it evaluates the estimator in a reproducible simulation study organised according to the ADEMP framework (Morris et al., 2019), using independent nodewise smoothing as the primary comparator and pooled smoothing and raw observations as additional reference methods.

The remainder of the thesis introduces the theoretical background in Section 2, defines the estimator and comparators in Section 3, and describes the empirical design in Section 4. Sections 5–8 present the results, discuss their implications, and conclude the thesis.

2 Theoretical Background

This chapter introduces the concepts needed to formulate the estimator in Section 3. The central idea is to combine two sources of regularity: smoothness of each function over its argument and similarity of functions attached to adjacent graph nodes. This places the thesis at the intersection of functional data analysis and spatially or network-indexed functional data analysis (Delicado et al., 2010). The graph is treated as known structural information; learning the graph from the observed curves is not part of the present problem.

2.1 Functional observations

In functional data analysis, the elementary observation is a function rather than a scalar or a finite-dimensional vector (Ramsay and Silverman, 2005). Let $Y_i : \mathcal{T} \rightarrow \mathbb{R}$ denote the functional observation for unit i , where $\mathcal{T} = [a, b]$ is a compact interval. Conceptually, the latent signal f_i is an element of $L^2(\mathcal{T})$, so that $\int_{\mathcal{T}} f_i(t)^2 dt < \infty$. In practice, only noisy evaluations $Y_i(t_{ij})$ are available. The present work considers a common and dense grid $t_1, \dots, t_T$ for all units, which permits the data to be stored as an $N \times T$ matrix without interpolation. This common-grid assumption is a restriction of the implementation used in this thesis, rather than a defining requirement of functional data analysis.

Treating the observations as functional is useful because neighbouring time points are not unrelated measurements. Regularisation over t can distinguish a systematic curve from observation noise, while the known graph provides a second direction in which the estimator can borrow information.

The formal observation model, reconstruction target and working-error assumptions used in this thesis are specified in Section 3.1.

2.2 Basis representations and P-splines

Following the standard basis-expansion approach to functional data and smooth regression (Ramsay and Silverman, 2005; Wood, 2017), a smooth function can be represented approximately in a finite-dimensional basis. For node i ,

$$f_i(t) \approx \sum_{k=1}^{K_t} \beta_{ik} b_k(t) = \mathbf{b}(t)^\top \boldsymbol{\beta}_i, \quad (1)$$

where $b_1, \dots, b_{K_t}$ are basis functions and $\boldsymbol{\beta}_i = (\beta_{i1}, \dots, \beta_{iK_t})^\top$. Evaluating the basis at the observation grid gives the matrix

$$B_{jk} = b_k(t_j), \quad j = 1, \dots, T, \quad k = 1, \dots, K_t. \quad (2)$$

The basis dimension determines the space in which the estimate can vary. A basis that is too small creates approximation bias that cannot be removed by changing a smoothing parameter, whereas an adequately rich basis can be regularised to prevent overfitting.

The implementation uses P-splines, which combine a B-spline basis with a difference penalty on neighbouring basis coefficients (Eilers and Marx, 1996). If D_q denotes the matrix that forms

differences of order q , the penalty matrix is

$$P_t = D_q^\top D_q, \quad \beta_i^\top P_t \beta_i = \|D_q \beta_i\|_2^2. \quad (3)$$

The spline degree and the difference-penalty order are separate modelling choices. The particular basis dimensions and penalty settings used in this thesis are specified with the fitting model in Section 3.5.

For scalar observations collected in $\mathbf{y}$ and design matrix $\mathbf{X}$, the penalised least-squares estimator is

$$\hat{\beta}_\lambda = \arg \min_{\beta} \left\{ \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \beta^\top P_t \beta \right\}. \quad (4)$$

Conditional on λ and assuming the relevant inverse exists,

$$\hat{\beta}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda P_t)^{-1} \mathbf{X}^\top \mathbf{y}. \quad (5)$$

A small λ prioritises fidelity to the data, while a large λ suppresses directions penalised by P_t . Because a difference penalty is rank deficient, functions in its null space remain unpenalised. With a first-order difference penalty, constant coefficient sequences form the null space. Identifiability constraints separate these unpenalised components from other model terms.

2.3 Smoothing parameters and REML

The smoothing parameter controls the bias–variance trade-off but should not be confused with the basis dimension. Increasing the basis dimension enlarges the available function space; increasing λ reduces the effective complexity within that space. For a Gaussian model, penalised spline coefficients can also be viewed as Gaussian random effects, with the penalty acting as a precision matrix. This mixed-model representation permits smoothing parameters to be estimated as variance-component ratios.

The models in this thesis use restricted maximum likelihood (REML). In broad terms, REML selects smoothing parameters by maximising a marginal likelihood after accounting for unpenalised fixed-effect components. It therefore avoids selecting the parameters by direct reference to the unknown true functions. For general smooth models with several penalties, stable estimation can be performed using derivatives of a Laplace-approximate marginal likelihood (Wood et al., 2016; Wood, 2017). The resulting smoothing parameters are data-adaptive and are estimated jointly, so the selected value for one penalty depends on the other terms and penalties in the model.

2.4 Graphs and the graph Laplacian

Let $G = (V, E)$ be an undirected, unweighted graph with $N = |V|$ nodes. Its symmetric adjacency matrix $A = (a_{ij})$ satisfies $a_{ij} = 1$ when nodes i and j are connected and $a_{ij} = 0$ otherwise. Self-edges are excluded, so $a_{ii} = 0$. The degree of node i is $d_i = \sum_j a_{ij}$, the degree matrix is $D = \text{diag}(d_1, \dots, d_N)$, and the unnormalised graph Laplacian (Shuman et al., 2013; Wood, 2017) is

$$L = D - A. \quad (6)$$

This is the Laplacian constructed by the MRF smoother from the neighbourhood list used in the thesis. A weighted extension replaces a_{ij} by a non-negative edge weight, but the current software interface uses binary neighbourhoods.

The matrix L is symmetric and positive semidefinite. For a vector of node-level values $\theta = (\theta_1, \dots, \theta_N)^\top$,

$$\theta^\top L \theta = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N a_{ij} (\theta_i - \theta_j)^2 \quad (7)$$

$$= \sum_{\{i,j\} \in E} (\theta_i - \theta_j)^2. \quad (8)$$

Consequently, the Laplacian quadratic form measures variation across graph edges, a standard measure of graph-signal roughness (Shuman et al., 2013). Penalising it shrinks connected nodes towards one another but does not directly shrink the overall level of θ .

The dimension of the null space of L equals the number of connected components. For a connected graph, $L\mathbf{1}_N = \mathbf{0}$ and the null space is spanned by the constant vector. A graph penalty alone therefore cannot identify an overall level. In a regression model this is handled by centering the graph effect and representing the common level separately.

2.5 Markov random-field smoothing

An MRF smooth uses the graph Laplacian as a quadratic penalty for a factor whose levels correspond to graph nodes (Wood, 2017). In a frequentist penalised-regression formulation, the contribution of a node effect θ is

$$\lambda_g \theta^\top L \theta, \quad (9)$$

where $\lambda_g \geq 0$ controls graph-based borrowing. The same quadratic structure appears in intrinsic conditional autoregressive models and has been applied to basis coefficients of spatially indexed curves (Zhang et al., 2016). The present estimator uses it only as a frequentist roughness penalty whose strength is estimated by REML. Because L is singular, the graph effect is centred and the common functional level is represented separately.

The quality of the graph matters. Equation (8) shrinks every linked pair, whether or not their true curves are similar. A correctly informative graph can reduce variance by pooling related nodes; an uninformative or misspecified graph can introduce bias. This motivates the simulation factors concerning graph-signal strength and neighbourhood definition.

2.6 Tensor-product smooths for graph-indexed functions

Graph-indexed functional data vary over the product domain $V \times \mathcal{T}$. A tensor-product construction combines marginal bases and marginal penalties for the graph and temporal domains (Wood, 2017; Scheipl et al., 2015). With one indicator basis function per node and a temporal basis of dimension K_t , the node-specific deviation can be written as

$$g(i, t) = \sum_{r=1}^N \sum_{k=1}^{K_t} \beta_{rk} \mathbb{1}\{i = r\} b_k(t). \quad (10)$$

There are two marginal notions of roughness: variation over the functional domain and variation across graph edges. The temporal component controls the complexity of node-specific curves over time, whereas the graph component encourages deviations at linked nodes to be similar. Tensor-product smooths retain separate regularisation parameters because these two forms of regularity act on different domains (Wood, 2017).

The functional additive modelling framework represents functional responses as additive models evaluated on the response grid (Scheipl et al., 2015; Greven and Scheipl, 2017). It provides the general modelling framework for combining a functional mean with graph-indexed deviations. The exact penalised criterion, centring constraint, and `pffr` implementation used in this thesis are specified in Section 3.

3 Methods

This chapter defines the reconstruction problem, the proposed estimator and its software implementation. The simulation design used to evaluate the estimator is described separately in Section 4.

3.1 Statistical setting and target

Suppose that one noisy curve is observed at every node of a known undirected, unweighted graph $G = (V, E)$ with binary adjacency matrix A . All curves are evaluated at the common grid $a \leq t_1 < \dots < t_T \leq b$. This yields an $N \times T$ response matrix, with one response-grid evaluation for every node–time pair. For node $i = 1, \dots, N$ and time point $j = 1, \dots, T$, the observation model is

$$Y_i(t_j) = f_i(t_j) + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2). \quad (11)$$

The Gaussian independent-error assumption is the working model used for estimation and in the core simulation. It is not intended to assert that all functional datasets have independent measurement errors. The graph, its node labels, and its neighbourhood relation are treated as fixed and known.

The target is the collection

$$\mathcal{F} = \{f_i : i = 1, \dots, N\} \quad (12)$$

at the observed graph nodes. Every target node therefore contributes data to the fit; prediction at previously unobserved nodes is not considered.

3.2 Functional mean and node-specific deviations

The curves are decomposed as

$$f_i(t) = \mu(t) + g_i(t), \quad (13)$$

where $\mu(t)$ is a functional mean and $g_i(t)$ is the deviation for node i . To distinguish the mean from the deviations, the interaction is centred over the graph nodes,

$$\sum_{i=1}^N g_i(t) = 0 \quad \text{for every } t \in \mathcal{T}. \quad (14)$$

In the fitted GAM this separation is achieved by the marginal centring constraints used for the tensor interaction. It is important to retain the functional intercept: removing it would force the graph-by-time term to represent both the common curve and the node deviations, conflicting with the centred interaction parameterisation.

Let $\mathbf{b}_0(t)$ be the K_0 -dimensional basis for the functional mean and $\mathbf{b}_1(t)$ the K_1 -dimensional temporal basis for the deviations. Then

$$\mu(t) = \mathbf{b}_0(t)^\top \boldsymbol{\alpha}, \quad g_i(t) = \mathbf{b}_1(t)^\top \boldsymbol{\beta}_i. \quad (15)$$

Stacking $\boldsymbol{\beta}_i$ by node gives

$$\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_N^\top)^\top \in \mathbb{R}^{NK_1}. \quad (16)$$

3.3 Penalised network estimator

Let X_0 and X_1 denote the design matrices for the functional mean and the centred node-by-time interaction, respectively. The rows correspond to all $N \times T$ node–time pairs (i, t_j) . Up to the constraint reparameterisation performed by the software, the proposed estimator minimises

$$\begin{aligned} (\hat{\boldsymbol{\alpha}}, \hat{\boldsymbol{\beta}}) = \arg \min_{\boldsymbol{\alpha}, \boldsymbol{\beta}} \bigg\{ & \|\mathbf{y} - X_0 \boldsymbol{\alpha} - X_1 \boldsymbol{\beta}\|_2^2 \\ & + \lambda_0 \boldsymbol{\alpha}^\top P_0 \boldsymbol{\alpha} \\ & + \lambda_t \boldsymbol{\beta}^\top (I_N \otimes P_1) \boldsymbol{\beta} \\ & + \lambda_g \boldsymbol{\beta}^\top (L \otimes I_{K_1}) \boldsymbol{\beta} \bigg\}. \end{aligned} \quad (17)$$

Here, P_0 and P_1 are temporal difference-penalty matrices. Let $D = \text{diag}(A\mathbf{1})$ be the degree matrix and $L = D - A$ the unnormalised graph Laplacian used by the MRF penalty (Shuman et al., 2013). The three smoothing parameters have distinct roles:

- λ_0 controls the temporal smoothness of the common mean curve;
- λ_t controls temporal smoothness of the node-specific deviations;
- λ_g controls similarity of deviations at adjacent nodes.

All three are estimated jointly from the same data by REML (Wood et al., 2016; Wood, 2017).

The graph component can be rewritten as

$$\boldsymbol{\beta}^\top (L \otimes I_{K_1}) \boldsymbol{\beta} = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N a_{ij} \|\boldsymbol{\beta}_i - \boldsymbol{\beta}_j\|_2^2 \quad (18)$$

$$= \sum_{\{i,j\} \in E} \|\boldsymbol{\beta}_i - \boldsymbol{\beta}_j\|_2^2. \quad (19)$$

Thus, the estimator borrows information by shrinking the complete temporal coefficient vectors of linked nodes towards each other. It does not merely smooth a scalar node intercept.

The limiting cases clarify the role of the graph penalty. Setting $\lambda_g = 0$ removes direct graph coupling, but the resulting joint model is not identical to the nodewise comparator in Section 3.7:

it still uses a common functional mean, a centred interaction and one shared temporal smoothing parameter for the interaction. For a connected graph, letting λ_g become very large forces adjacent coefficient vectors towards equality. Combined with the centring constraint, this suppresses the node deviations and leaves the common mean curve. Hence the fitted model provides data-adaptive partial pooling between heterogeneous node curves and a common curve.

3.4 Tensor-product representation in `pffr`

The estimator is expressed as a function-on-scalar regression using the functional additive mixed-model framework of Scheipl et al. (2015) and the general functional-regression formulation of Greven and Scheipl (2017). In the user-facing formula, the functional response is regressed on an MRF smooth of the node factor:

```
Y ~ s(
  node,
  bs = "mrf",
  xt = list(nb = nb_list),
  k = N
)
```

The formula deliberately contains neither a `c()` wrapper nor `-1`. In `pffr`, a term wrapped in `c()` would represent an effect that is constant over the response index; this would permit only node-specific levels rather than node-specific curve shapes. Likewise, removing the intercept would eliminate the required functional mean.

Internally, the specification is expanded conceptually as

```
s(yindex.vec) + ti(node, yindex.vec, ...)
```

The first term estimates $\mu(t)$. The second is a pure graph-by-time interaction constructed from an MRF marginal for `node` and a P-spline marginal for the response index. The `ti` construction applies marginal centring constraints and associates two penalties with the interaction: one in the graph direction and one in the temporal direction. This corresponds to the last two penalty terms in Equation (17), after the internal basis transformation used for identifiability.

3.5 Fitting specification used in the analyses

The network-weighted estimator uses a Gaussian identity link and estimates its three smoothing parameters jointly by REML using all $N \times T$ response-grid evaluations simultaneously. The basis specifications used for this estimator and for the comparator methods were fixed before the main simulation. Following the P-spline construction introduced in Section 2.2, the functional intercept is represented by cubic B-spline basis functions with a first-order difference penalty and basis dimension $K_0 = 10$:

```
bs.int = list(bs = "ps", k = 10, m = c(2, 1))
```

The temporal marginal of the node-by-time deviation uses the same cubic B-spline basis and first-order difference penalty with $K_1 = 15$:

```
bs.yindex = list(bs = "ps", k = 15, m = c(2, 1))
```

In the `mgcv` P-spline parameterisation, `m = c(2, 1)` therefore encodes cubic B-spline basis functions and a first-order difference penalty on adjacent coefficients (Eilers and Marx, 1996; Wood, 2017). The graph marginal uses `k = N`, so no low-rank truncation of the set of node levels is imposed.

The choice $K_1 = 15$ was checked in the pilot by projecting the true deviations into the unpenalised temporal basis. This check confirmed that $K_1 = 15$ could represent the true deviations sufficiently well, so that the comparison between estimators was not driven by an inadequate temporal basis dimension. Supplementary implementation details concerning graph conversion and graph–curve identifier alignment are provided in Appendix A.1.

3.6 Graph representation and node alignment

The analyses use binary, undirected graphs. In the named neighbourhood-list workflow implemented here, an adjacency matrix records only the presence or absence of an edge: non-zero off-diagonal entries are treated as edges, and their magnitudes are not retained. This is an implementation choice for the present workflow rather than a general restriction of `mgcv`'s MRF facilities. The `netfunsmooth` implementation converts the supplied graph to the named neighbourhood-list representation required by the MRF marginal, whose names agree with the levels of the node factor. Before fitting, curve identifiers and graph-node identifiers are checked for uniqueness and exact agreement, after which the curves are reordered to graph order. Predictions retain this ordering.

This alignment step is part of the statistical implementation rather than a cosmetic data operation. The MRF penalty is indexed by factor levels, so matching dimensions alone is insufficient: permuting the curve rows while leaving the graph labels unchanged would associate observations with the wrong neighbourhoods and therefore change the fitted estimator. Supported graph and functional-data representations, their validation rules, and polygon-identifier handling are described in Appendix A.1.

3.7 Comparator estimators

The simulation uses three benchmarks to separate the effects of temporal smoothing and graph-based pooling.

Raw observation No model is fitted and

$$\hat{f}_i(t_j) = Y_i(t_j). \quad (20)$$

This descriptive reference quantifies the error before any temporal or graph smoothing.

Nodewise smoothing Every node is fitted separately using

```
mgcv::gam(
```

```

y ~ s(t, bs = "ps", k = 15, m = c(2, 1)),
family = gaussian(),
method = "REML"
)

```

The spline basis, penalty and basis dimension match the temporal marginal of the network deviation. Each node nevertheless receives its own REML-selected smoothing parameter. This is the primary reconstruction comparator. The comparison assesses whether the complete network-weighted estimator improves reconstruction accuracy beyond independent nodewise smoothing. Because the network estimator also contains a common functional mean and jointly selected smoothing parameters, this comparison alone does not isolate the contribution of the particular supplied graph edges.

Pooled mean smoothing Node identity is discarded and one curve is fitted to all $N \times T$ response-grid evaluations.

```

mgcv::gam(
  y ~ s(t, bs = "ps", k = 10, m = c(2, 1)),
  family = gaussian(),
  method = "REML"
)

```

The estimated mean is assigned to every node. This represents complete pooling and uses the same temporal specification as the functional intercept of the network model. It is a diagnostic benchmark for the cost of ignoring node-specific variation, rather than the primary competitor.

Within a simulated dataset, all methods receive the same observations and are evaluated on the same grid. Functional kriging is not included in the primary comparison because the present estimand is denoising and reconstruction at observed graph nodes. Including it would require a separate node-holdout prediction design and would therefore address a different estimand. The restricted-oracle analysis is a simulation-only diagnostic for smoothing-parameter selection and is not an applicable estimator for real data because it uses the known truth.

3.8 Scope of the method

The proposed estimator is intended for settings in which a supplied binary, undirected graph encodes a scientifically meaningful notion of local similarity. This is a modelling assumption rather than a property learned from the data. The penalty does not prohibit discontinuities, but it penalises differences across graph edges; consequently, genuine abrupt boundaries may be oversmoothed.

A single global graph smoothing parameter controls the overall strength of graph-based borrowing. The model does not estimate edge-specific or locally adaptive borrowing strengths. In the implementation used here, a non-zero off-diagonal entry of an adjacency matrix defines an unweighted neighbourhood relation, so edge magnitudes are not retained.

This thesis considers one noisy Gaussian curve at each observed node, with all curves evaluated on a common functional grid. The common grid need not be equally spaced, but node-

specific irregular or sparse observation schemes are not considered. The target is reconstruction at the observed graph nodes; prediction at previously unobserved nodes is outside the present scope. Extensions to weighted or directed graphs, locally adaptive penalties, non-Gaussian functional responses, and more general observation designs remain topics for future work.

4 Simulation Study

The empirical evaluation is organised according to the ADEMP framework of Morris et al. (2019): aims, data-generating mechanisms, estimands, methods, and performance measures. The data-generating mechanisms, primary comparison, and Monte Carlo rules were fixed after the design audit and pilot, but before the 200-replication main results were inspected. The primary analysis is the 16-cell core experiment described below. A separate $\alpha = 0$ zero-structured-signal diagnostic examines the behaviour of the estimator when the supplied graph is not deliberately aligned with the node-specific deviations. The additional diagnostics described below are reported separately and do not determine the primary conclusion.

4.1 Aims

The primary aim is to assess whether network-weighted smoothing reconstructs node-specific curves more accurately than independent nodewise smoothing when neighbouring nodes carry related functional signals. The simulation also examines how this comparison changes with four design features: the spatial structure of the truth, the strength of its structured component, the observation-noise level, and the neighbourhood graph supplied to the estimator. Raw observations and complete pooling are included as descriptive reference points.

The estimand is the collection of true curves at the observed graph nodes,

$$\mathcal{F} = \{f_1, \dots, f_N\}. \quad (21)$$

Thus, the study evaluates reconstruction at observed nodes. It does not evaluate prediction of an entire curve at an unobserved spatial location; functional kriging would require a separate node-holdout design and is outside the scope of this thesis.

4.2 Data-generating mechanisms

4.2.1 Graph and observation model

The simulation uses a regular 5×5 lattice with $N = 25$ nodes. Each curve is observed at $T = 50$ equally spaced points $t_j \in [0, 1]$. The observations are generated from

$$Y_i(t_j) = f_i(t_j) + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad (22)$$

for $i = 1, \dots, N$ and $j = 1, \dots, T$. The two noise levels are $\sigma \in \{0.2, 0.5\}$, representing a comparatively low- and high-noise setting. Gaussian independent errors deliberately provide

a simple reference case in which differences between methods can be attributed to smoothing and the use of graph information.

Two neighbourhood definitions are supplied to the network estimator. Under rook adjacency, nodes are neighbours when their grid cells share an edge; the resulting graph contains 40 edges. Queen adjacency additionally connects cells that share a corner and contains 72 edges. Importantly, these graphs are not used to generate the truth. For each combination of the remaining factors and each replication, the same true curves and noisy observations are used for both graph fits. The rook–queen comparison therefore changes only the neighbourhood information available to the estimator.

4.2.2 True curves and signal strength

Each node-specific curve is decomposed into a common mean and a centred deviation,

$$f_i(t) = \mu(t) + \delta_i(t), \quad (23)$$

$$\mu(t) = 1 + 0.5 \sin(2\pi t) + 0.25 \cos(4\pi t), \quad (24)$$

$$\delta_i(t) = \sum_{k=1}^3 \theta_{ik} \phi_k(t), \quad \sum_{i=1}^N \delta_i(t) = 0, \quad (25)$$

where

$$\phi_1(t) = \sin(2\pi t), \quad \phi_2(t) = \cos(2\pi t), \quad \phi_3(t) = \exp\{-100(t - 0.65)^2\}. \quad (26)$$

The three components allow node curves to differ in more than their overall level: the varying coefficients produce changes in amplitude, phase, and the height of a local peak. This is important because the fitted model contains a node-by-time interaction and should be assessed on its ability to reconstruct differences in curve shape.

For temporal component k , the coefficient vector is constructed as

$$\boldsymbol{\theta}_k = s_k \left[\sqrt{\alpha} \boldsymbol{\theta}_k^{(S)} + \sqrt{1 - \alpha} \boldsymbol{\theta}_k^{(U)} \right]. \quad (27)$$

For each temporal component, the structured and unstructured fields are centred and scaled to unit average squared magnitude across nodes; the unstructured field is also made orthogonal to its structured counterpart before they are combined. Consequently, α is the proportion of across-node coefficient variation contributed by the structured component, while the marginal average squared magnitude remains fixed. The scale s_k is the root mean square of the centred coordinate-smooth field for component k and is used for both truth structures. It therefore keeps the marginal coefficient scales comparable between the two data-generating mechanisms. The core experiment uses $\alpha \in \{0.5, 0.9\}$, corresponding to moderate and strong structured signals.

Two forms of $\boldsymbol{\theta}_k^{(S)}$ are considered. In the *coordinate-smooth* data-generating mechanism, let x_i and y_i denote centred and range-normalised lattice coordinates in $[-1, 1]$, where x_i indexes the lattice column and y_i the lattice row. The three raw fields are

$$h_{i1} = 0.8x_i, \quad h_{i2} = 0.8y_i, \quad h_{i3} = 0.6x_iy_i. \quad (28)$$

Their centred and standardised versions define the structured coefficient fields. They induce gradual variation across the lattice and represent a case in which borrowing information from nearby nodes should be beneficial.

In the *cluster-structured* mechanism, the first two lattice columns form one connected cluster and the final three columns form a second. The raw cluster contrasts for the three temporal components are

$$\mathbf{\Gamma} = \begin{pmatrix} 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}. \quad (29)$$

Each node receives the row of $\mathbf{\Gamma}$ corresponding to its cluster, followed by the same centring and standardisation used above. This mechanism is piecewise smooth but discontinuous at the cluster boundary. It tests whether graph regularisation can reduce noise without excessively blurring a genuine boundary.

For both mechanisms, $\theta_k^{(U)}$ is generated from an independent standard-normal vector, residualised with respect to the intercept and the structured component, and rescaled to unit average squared magnitude across nodes. The same realisation is reused across the paired graph fits. Hence neither truth is defined through eigenvectors of the fitted graph, which avoids changing the estimand when the neighbourhood definition changes.

4.2.3 Factorial design and additional runs

The four factors produce the core design shown in Table 1.

Table 1: Factors in the core simulation study.

Factor	Levels
Truth structure	Coordinate-smooth; cluster-structured
Structured-signal proportion	$\alpha = 0.5$; $\alpha = 0.9$
Estimator graph	Rook; queen
Noise standard deviation	$\sigma = 0.2$; $\sigma = 0.5$

The resulting $2 \times 2 \times 2 \times 2 = 16$ cells are evaluated using paired random numbers wherever possible. Within a replication, all four methods receive the same dataset; rook and queen network fits share the same truth and observations; and the same standard-normal error array is scaled to construct the two noise levels. This pairing reduces Monte Carlo variation in the comparisons of interest.

Before the core experiment, a separate known-answer check uses the coordinate-smooth truth, $\alpha = 1$, the rook graph, and $\sigma = 0.5$. This setting is not included in the 16-cell analysis. Its purpose is to verify that the implementation detects a setting in which graph smoothing should clearly improve upon nodewise smoothing.

It uses $B_{\text{sanity}} = 50$ attempted replications and three prespecified pass criteria for the paired improvement defined in Equation (33): a positive mean improvement, a mean improvement

exceeding twice its paired Monte Carlo standard error, and an improvement in at least 80% of replications. The check is diagnostic rather than an additional estimate of primary performance.

A separate $\alpha = 0$ zero-structured-signal diagnostic complements the core design. It uses the coordinate-labelled construction with $\alpha = 0$, so that the deliberately structured coefficient component receives zero weight and the node-specific deviations are not deliberately aligned with the supplied graph. The diagnostic is evaluated at $\sigma = 0.5$ with 200 attempted replications for each estimator graph and is not included in the 16-cell core experiment.

A supplementary fixed rewired-graph control examines the role of the particular supplied edges in two selected rook-graph settings: coordinate-smooth and cluster-structured truth, both with $\alpha = 0.9$ and $\sigma = 0.5$. Within each paired replication, the true curves and observations are held fixed while the correct rook graph is compared with one connected degree-preserving rewired graph. This control is reported separately from the 16-cell core experiment in Section 5.6.1.

A prespecified computational-scalability timing gate assesses the node-count dimension separately from the core design. It considers two fixed $N = 100$ cells on a 10×10 rook lattice with $T = 50$ and $\sigma = 0.5$: the coordinate-smooth setting with $\alpha = 0.9$ and the coordinate-labelled zero-structured-signal setting with $\alpha = 0$. Each timing replication includes data generation, fitting both the network-weighted and nodewise estimators, prediction, and error calculation.

Before any $N = 100$ performance comparison is conducted, five complete timing replications are run for each cell. Their median complete-replication times are used to project the serial runtime for $B_{100} = 25$ attempted replications per cell. The $N = 100$ comparison is run only if this projection does not exceed the prespecified eight-hour serial budget. Otherwise, the timing gate is reported as a feasibility result only, and no smaller ad hoc replication count or $N = 100$ accuracy analysis is substituted.

4.3 Methods compared

Four reconstruction methods are applied to each simulated dataset.

1. **Raw observations** use $Y_i(t_j)$ directly and provide a no-smoothing benchmark.
2. **Pooled smoothing** ignores node identity, estimates one common P-spline from all observations, and assigns it to every node.
3. **Nodewise smoothing** fits an independent P-spline to each node using REML. This is the primary comparator because it uses temporal regularisation but no graph information.
4. **Network-weighted smoothing** is the estimator developed in Section 3. It jointly estimates the curves with a smooth functional intercept and a node-by-time interaction carrying temporal and graph-Laplacian penalties. Its smoothing parameters are selected by REML, following the functional regression and general smooth-model frameworks of Greven and Scheipl (2017) and Wood et al. (2016).

The P-spline basis specifications are fixed across replications. The functional intercept and pooled method use basis dimension $K_{\text{int}} = 10$; the temporal marginal of the interaction and each nodewise smooth use $K_{\text{dev}} = 15$. All use cubic B-splines with first-order difference penalties. Before the main run, the unpenalised projection error of the true deviations is checked to ensure

that the chosen basis does not impose a material approximation-error floor. The deterministic design audit passed before the main simulation. Across the 50 pilot truth draws and four core truth configurations, the maximum relative unpenalised L^2 projection error for the node-specific deviations was 0.412%, below the prespecified threshold of 2%.

4.4 Performance measures

For node i , replication b , design cell c , and method m , reconstruction error is quantified by integrated squared error,

$$\text{ISE}_{ibcm} = \int_0^1 \left\{ \hat{f}_{ibcm}(t) - f_{ibc}(t) \right\}^2 dt. \quad (30)$$

The integral is approximated on the common 50-point time grid using trapezoidal weights. Within a replication, node-specific errors are averaged as

$$\text{AISE}_{bcm} = \frac{1}{N} \sum_{i=1}^N \text{ISE}_{ibcm}. \quad (31)$$

Let $\mathcal{S}_{cm}$ denote the set of replications in which method m successfully fitted in cell c , and let $n_{cm} = |\mathcal{S}_{cm}|$. The primary absolute measure is the Monte Carlo mean of AISE_{bcm} ,

$$\widehat{\text{MISE}}_{cm} = \frac{1}{n_{cm}} \sum_{b \in \mathcal{S}_{cm}} \text{AISE}_{bcm}. \quad (32)$$

The primary network-versus-nodewise comparison is paired within the same simulated dataset:

$$\Delta_{bc} = \text{AISE}_{bc, \text{nodewise}} - \text{AISE}_{bc, \text{network}}. \quad (33)$$

Positive values favour network-weighted smoothing. The mean paired improvement and its Monte Carlo standard error are calculated only from the jointly successful set $\mathcal{S}_c^{(\text{pair})}$, containing replications in which both network and nodewise fits succeed. Writing $n_c^{(\text{pair})} = |\mathcal{S}_c^{(\text{pair})}|$, these are

$$\hat{\Delta}_c = \frac{1}{n_c^{(\text{pair})}} \sum_{b \in \mathcal{S}_c^{(\text{pair})}} \Delta_{bc}, \quad (34)$$

$$\text{MCSE}(\hat{\Delta}_c) = \frac{\text{SD} \left\{ \Delta_{bc} : b \in \mathcal{S}_c^{(\text{pair})} \right\}}{\sqrt{n_c^{(\text{pair})}}}. \quad (35)$$

The replication-wise win rate is the proportion of jointly successful replications in which network-weighted smoothing has lower node-averaged ISE:

$$\hat{p}_{\text{win},c} = \frac{1}{n_c^{(\text{pair})}} \sum_{b \in \mathcal{S}_c^{(\text{pair})}} \mathbf{1}\{\Delta_{bc} > 0\}. \quad (36)$$

The indicator equals one when network-weighted smoothing has lower node-averaged ISE than nodewise smoothing in replication b , and equals zero otherwise.

For this same jointly successful set, the paired MISE ratio is

$$R_c^{\text{MISE}} = \frac{\sum_{b \in \mathcal{S}_c^{(\text{pair})}} \text{AISE}_{bc,\text{network}}}{\sum_{b \in \mathcal{S}_c^{(\text{pair})}} \text{AISE}_{bc,\text{nodewise}}}. \quad (37)$$

Values below one favour network-weighted smoothing. The corresponding percentage, $100(1 - R_c^{\text{MISE}})$, is reported as the relative MISE reduction versus nodewise smoothing. This aggregate ratio is preferred to a mean of replication-specific relative errors because the latter has a random denominator and can be unstable.

Failure rates and elapsed computation times are reported alongside reconstruction accuracy. Median-node and worst-node ISE are retained as secondary descriptive summaries, but do not determine the main conclusion.

4.4.1 Cluster-boundary diagnostic

For the cluster-structured truth, a prespecified secondary diagnostic examines whether the network estimator's advantage changes at the discontinuity. The matched boundary set is the third lattice column (nodes 11–15), and the matched interior set is the fourth column (nodes 16–20). Both sets contain five nodes in the same cluster and have the same row positions and graph degrees under each neighbourhood definition. For node i , define the network-relative error as

$$d_{ibc} = \text{ISE}_{ibc,\text{network}} - \text{ISE}_{ibc,\text{nodewise}}. \quad (38)$$

The boundary difference-in-differences is the mean of d_{ibc} over boundary nodes minus its mean over matched interior nodes. Positive values indicate that network smoothing has an additional boundary-specific loss relative to nodewise smoothing, or equivalently, that its advantage is attenuated at the boundary. This diagnostic is summarised over jointly successful replications with its paired MCSE; it does not replace the primary all-node comparison.

4.4.2 Restricted-oracle tuning diagnostic

A restricted-oracle diagnostic examines the gap between REML smoothing-parameter selection and the best selection available within a fixed candidate range. It is evaluated in four prespecified rook-graph cells: coordinate-smooth truth with $\alpha = 0.9$ and $\sigma \in 0.2, 0.5$, cluster-structured truth with $\alpha = 0.9$ and $\sigma = 0.5$, and the coordinate-labelled $\alpha = 0$ zero-structured-signal diagnostic with $\sigma = 0.5$. The diagnostic uses $B_{\text{oracle}} = 50$ paired replications per cell.

For each simulated dataset, the network model is first fitted by REML. The functional-intercept smoothing parameter is then fixed at its REML value. The graph and time interaction smoothing parameters are varied on a 5×5 coarse grid of log-scale offsets $-4, -2, 0, 2, 4$ relative to their REML values. A 3×3 local refinement with offsets $-1, 0, 1$ around the best coarse candidate is then evaluated, with duplicate candidates omitted. The zero-offset candidate is included as an implementation check because it reproduces the REML fit.

The oracle selects the candidate with the lowest AISE relative to the known true curves. It is therefore a simulation-only tuning diagnostic rather than a practically available competing estimator. Paired REML-minus-oracle AISE differences, their paired MCSEs, and oracle win rates are reported descriptively. For each cell, the frequency with which a successful coarse search selects a graph or time boundary is also recorded. A frequency above 5% classifies the cell as range-inadequate; such a cell is not used for formal oracle-comparison conclusions, and its candidate range is not expanded post hoc.

4.5 Monte Carlo design and reproducibility

The replication count for the core Monte Carlo experiment was determined from a pilot with $B_{\text{pilot}} = 50$ attempted replications per core cell. Writing $\bar{\Delta}_c^{\text{pilot}}$ and $s_{\Delta,c}^{\text{pilot}}$ for the pilot mean and standard deviation of the paired differences in core cell c , the prespecified tolerance was

$$\tau_c = \max \left\{ 0.10 \left| \bar{\Delta}_c^{\text{pilot}} \right|, 0.01 \widehat{\text{MISE}}_{c,\text{nodewise}}^{\text{pilot}} \right\}. \quad (39)$$

The estimated requirement for the main simulation in cell c was

$$B_c = \left\lceil \left(\frac{s_{\Delta,c}^{\text{pilot}}}{\tau_c} \right)^2 \right\rceil, \quad (40)$$

and the common replication count was

$$B = 50 \left\lceil \frac{\max\{200, \max_c B_c\}}{50} \right\rceil. \quad (41)$$

In the pilot, the largest estimated cell-specific requirement was $B_c = 31$. The floor of 200 attempts was therefore binding, and the main analysis used $B = 200$ paired replications for each graph-specific cell. Across the eight truth–signal–noise combinations, this produced 1,600 distinct simulated response datasets. Each response dataset was analysed under both the rook and queen graph specifications, yielding 3,200 graph-specific cell–replication evaluations. Pilot runs are not included in the final performance summaries.

Failed fits are recorded and are not replaced with newly generated datasets. Method-specific summaries use the corresponding successful fits, whereas paired comparisons use only jointly successful replications and report their actual sample size.

All randomness is generated from a frozen L'Ecuyer–CMRG stream chain. The pilot, formal known-answer check, and 16-cell core experiment use disjoint stream ranges: positions 1–50 for the pilot, 51–100 for the known-answer check, and 101–300 for the main run. The separate $\alpha = 0$ diagnostic deliberately reuses the main-run streams while applying its own fixed data-generating mechanism; its results are stored and analysed separately and are not combined

with the core summaries. Random-number streams, scenario identifiers, and replication-level outputs are retained so that interrupted runs can be resumed without changing the generated datasets.

5 Results

5.1 Simulation completion and implementation checks

The frozen core experiment was completed as specified in Section 4. It comprised 1,600 distinct simulated response datasets, generated from the $2 \times 2 \times 2$ combinations of truth structure, α , and noise level, with 200 replications each. Each dataset was evaluated under both rook and queen graph specifications, yielding 16 graph-specific design cells and 3,200 graph-specific cell–replication evaluations. The saved-output checks found the expected number of records, no duplicated cell–replication–method combinations, and 200 attempted replications in every graph-specific cell. The resulting output contains 12,800 stored method-specific result records; these are not 12,800 independent datasets. The primary network–nodewise comparison was jointly successful in all 200 replications of every design cell. Thus, no cell-specific accuracy comparison relies on a reduced set of successful paired fits.

The separate formal known-answer check described in Section 4.2.3 comprised 50 attempted replications. Both the nodewise and network-weighted fits were successful in all 50 replications. The mean paired improvement $\text{AISE}_{\text{nodewise}} - \text{AISE}_{\text{network}}$ was 0.009446, with a paired MCSE of 0.000272; network-weighted smoothing had lower AISE in 100% of replications. The check therefore passed all three prespecified performance criteria. It is reported as an implementation diagnostic and is not included in the 16-cell primary performance summaries.

These checks concern the numerical execution of the simulation rather than accuracy. In particular, the main conclusion below is based on the paired reconstruction-error comparison defined in Equation (33), not on convergence alone.

5.2 $N = 100$ scalability timing pilot

A prespecified computational-scalability timing gate considered two fixed $N = 100$ cells: a coordinate-smooth rook-lattice setting with $\alpha = 0.9$ and a coordinate-labelled unstructured zero-structured-signal setting with $\alpha = 0$. Both used a 10×10 rook lattice, $T = 50$ observation points, and $\sigma = 0.5$. As planned, this extension compared only network-weighted REML smoothing with separate nodewise smoothing; it was not part of the 16-cell core performance experiment.

Before any $N = 100$ performance summaries were produced, five complete timing replications were run for each cell. Each timing replication included data generation, fitting both methods, prediction, and error calculation. The median elapsed time was 1,288.60 seconds (21.5 minutes) for the $\alpha = 0.9$ cell and 2,343.86 seconds (39.1 minutes) for the $\alpha = 0$ cell. Applying the prespecified serial-runtime rule, even the smallest candidate size, $B_{100} = 25$ attempted replications per cell, was projected to require 90,811.5 seconds (25.2 hours), exceeding the eight-hour budget.

The $N = 100$ main block was therefore declared computationally infeasible and was not run. No smaller ad hoc replication count was substituted, and no $N = 100$ MISE, paired-improvement,

smoothing-parameter, or effective degrees of freedom (EDF) summaries are reported. Thus, this pilot provides a computational feasibility result only; it does not provide evidence about reconstruction accuracy at $N = 100$. The completed $N = 25$ core experiment remains the basis for the performance conclusions that follow.

5.3 Primary comparison: network versus nodewise smoothing

Network-weighted smoothing had lower estimated MISE than nodewise smoothing in all 16 core cells. Table 2 reports the MISE ratio and relative MISE reduction. Table 3 reports paired evidence across the 200 jointly successful replications per cell: paired improvement is nodewise minus network node-averaged reconstruction error, so positive values favour network-weighted smoothing; its paired MCSE quantifies Monte Carlo uncertainty, and the win rate is the proportion of replications in which the network-weighted estimator has lower node-averaged ISE. Figure 1 visualises the relative reduction across the factorial design.

The relative reduction ranged from 1.4% to 20.5%. In every cell, the mean paired improvement was positive by more than twice its paired Monte Carlo standard error, and the network-weighted estimator won in 72.0% to 100.0% of individual replications. These results support the primary conclusion that the complete joint estimator achieved lower reconstruction error than separate nodewise smoothing in all 16 core simulation settings. This comparison evaluates the full joint modelling approach: in addition to the graph penalty, the estimator includes a common functional mean and estimates its smoothing parameters jointly across nodes. It therefore does not by itself isolate the contribution of the particular supplied graph edges.

Table 2: Primary network-versus-nodewise comparison. MISE ratios below one favour network-weighted smoothing.

Truth	α	Graph	σ	MISE ratio	Reduction (%)
Cluster	0.5	Queen	0.2	0.954	4.58
Cluster	0.5	Queen	0.5	0.874	12.56
Cluster	0.5	Rook	0.2	0.955	4.48
Cluster	0.5	Rook	0.5	0.873	12.72
Cluster	0.9	Queen	0.2	0.984	1.64
Cluster	0.9	Queen	0.5	0.895	10.53
Cluster	0.9	Rook	0.2	0.986	1.43
Cluster	0.9	Rook	0.5	0.875	12.53
Coordinate	0.5	Queen	0.2	0.949	5.08
Coordinate	0.5	Queen	0.5	0.867	13.32
Coordinate	0.5	Rook	0.2	0.949	5.11
Coordinate	0.5	Rook	0.5	0.867	13.34
Coordinate	0.9	Queen	0.2	0.957	4.35
Coordinate	0.9	Queen	0.5	0.813	18.75
Coordinate	0.9	Rook	0.2	0.956	4.40
Coordinate	0.9	Rook	0.5	0.795	20.46

Table 3: Paired network-versus-nodewise comparison across 200 replications per cell.

Truth	α	Graph	σ	Mean paired improvement	Paired MCSE	Win rate (%)
Cluster	0.5	Queen	0.2	0.000321	0.000013	98.0
Cluster	0.5	Queen	0.5	0.004305	0.000125	99.5
Cluster	0.5	Rook	0.2	0.000314	0.000013	96.0
Cluster	0.5	Rook	0.5	0.004359	0.000126	99.5
Cluster	0.9	Queen	0.2	0.000114	0.000013	74.5
Cluster	0.9	Queen	0.5	0.003535	0.000154	94.5
Cluster	0.9	Rook	0.2	0.000099	0.000013	72.0
Cluster	0.9	Rook	0.5	0.004205	0.000158	99.0
Coordinate	0.5	Queen	0.2	0.000353	0.000012	97.5
Coordinate	0.5	Queen	0.5	0.004514	0.000122	100.0
Coordinate	0.5	Rook	0.2	0.000355	0.000013	97.5
Coordinate	0.5	Rook	0.5	0.004520	0.000127	99.0
Coordinate	0.9	Queen	0.2	0.000300	0.000014	96.0
Coordinate	0.9	Queen	0.5	0.006209	0.000134	100.0
Coordinate	0.9	Rook	0.2	0.000303	0.000015	96.0
Coordinate	0.9	Rook	0.5	0.006775	0.000140	100.0

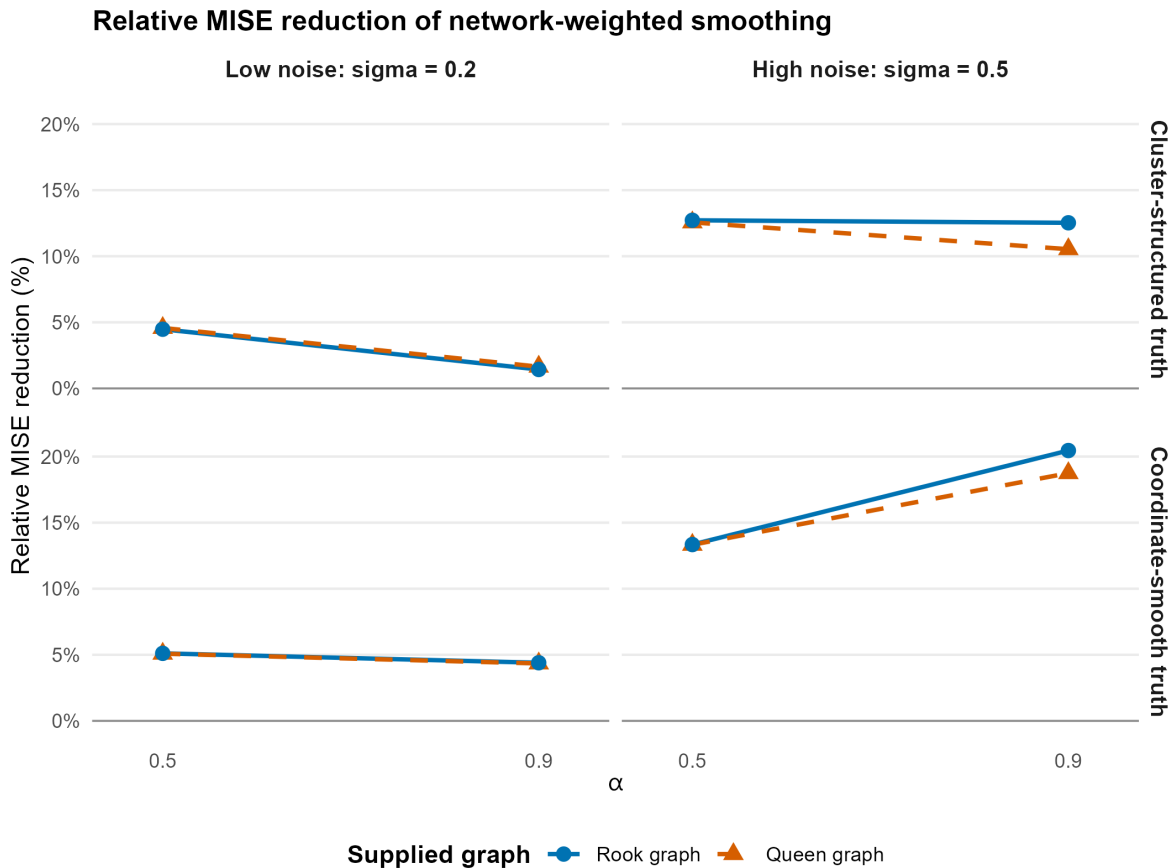


Figure 1: Relative MISE reduction of network-weighted smoothing relative to separate node-wise smoothing across the 16 core simulation cells. Positive values favour network-weighted smoothing. Lines distinguish the rook and queen neighbourhood graphs.

For context, the pooled reference method had the largest MISE in every setting, whereas the raw curves had larger MISE than both smoothed node-specific estimators; the corresponding absolute MISE values are reported in Appendix A.2.

5.4 Patterns across noise, graph signal, and neighbourhood density

The most pronounced pattern is the role of observation noise. At $\sigma = 0.2$, the relative MISE reductions were modest, between 1.4% and 5.1%. At $\sigma = 0.5$, they were larger in every corresponding design cell, ranging from 10.5% to 20.5%. This is the expected variance-reduction role of network regularisation: borrowing information across related nodes is most useful when each individual curve is noisier.

The effect of structured-signal strength depends on the form of the truth. For the coordinate-smooth truth at high noise, the gain increased from about 13.3% at $\alpha = 0.5$ to 18.7%–20.5% at $\alpha = 0.9$. In contrast, the cluster-structured truth contains a genuine discontinuity. At low noise, the gain was smaller when its structured component was stronger (about 1.4%–1.6% at $\alpha = 0.9$), consistent with a more difficult bias–variance trade-off near the boundary. Even there, the all-node comparison continued to favour the network-weighted estimator.

No uniform ordering of rook and queen neighbourhoods was observed. Their results were almost identical in several cells. The largest difference occurred for cluster-structured truth with $\alpha = 0.9$ and $\sigma = 0.5$, where the rook graph gave a 12.5% reduction and the queen graph a 10.5% reduction. Therefore, the core experiment supports an advantage of the network-weighted estimator in these settings, but it does not support a general rule that the denser queen graph is always preferable.

5.5 Cluster-boundary diagnostic

The prespecified boundary difference-in-differences analysis was conducted for the eight cluster-structured cells. For each replication, the diagnostic compares the network-minus-nodewise reconstruction-error difference at the cluster boundary with the corresponding difference at matched interior nodes. A positive Boundary DiD therefore indicates that the network-weighted estimator’s advantage is smaller at the boundary than in the interior.

Table 4 reports the cell-specific estimates. At $\sigma = 0.2$, the Boundary DiD estimates ranged from 0.000048 to 0.000060, with paired MCSEs between 0.000046 and 0.000049. Thus, all four two-MCSE intervals included zero. At $\sigma = 0.5$, the estimates ranged from 0.001216 to 0.002750, with paired MCSEs between 0.000487 and 0.000516. All four corresponding two-MCSE intervals lay above zero.

Table 4: Boundary difference-in-differences for cluster-structured truth.

α	σ	Graph	Boundary DiD	Paired MCSE
0.5	0.2	Queen	0.000055	0.000049
0.5	0.2	Rook	0.000048	0.000049
0.9	0.2	Queen	0.000060	0.000046
0.9	0.2	Rook	0.000051	0.000047
0.5	0.5	Queen	0.001459	0.000516
0.5	0.5	Rook	0.001216	0.000514
0.9	0.5	Queen	0.002750	0.000487
0.9	0.5	Rook	0.002313	0.000498

Figure 2 visualises the same estimates. Points show the mean Boundary DiD and error bars show plus or minus two paired MCSEs. The vertical scale is multiplied by 10^{-3} . The figure makes clear that the boundary-specific attenuation is not apparent at low noise but is positive in all four high-noise cells.

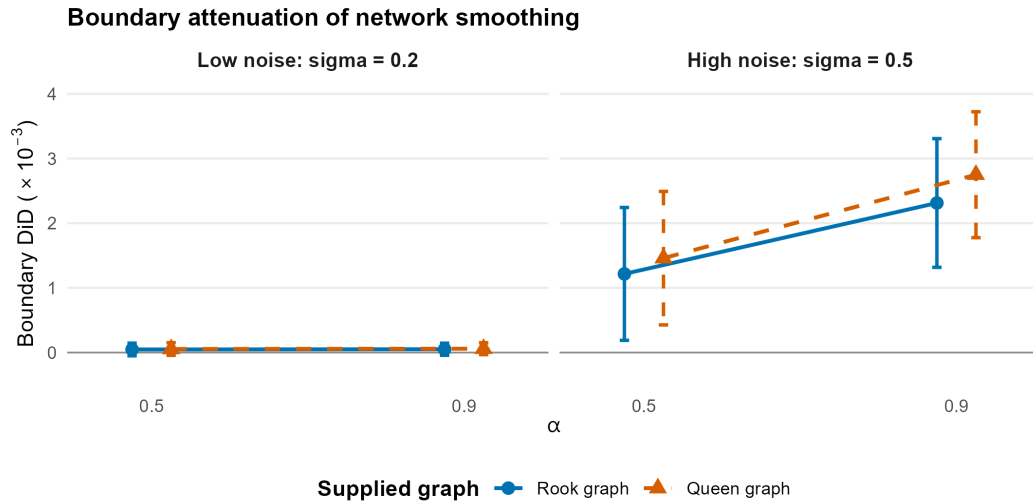


Figure 2: Boundary difference-in-differences for cluster-structured truth; error bars show plus or minus two paired MCSEs.

The positive high-noise Boundary DiD values should not be interpreted as showing that network smoothing is worse than nodewise smoothing at boundary nodes. In all four high-noise cells, the mean boundary reconstruction-error difference, defined as network minus nodewise error, remained negative and ranged from -0.003154 to -0.000909 . Thus, network smoothing still reduced average boundary-node error. Its improvement was nevertheless attenuated at the discontinuity, particularly when $\alpha = 0.9$. This pattern is consistent with the anticipated boundary-specific bias-variance trade-off of graph regularisation.

5.6 Zero-structured-signal diagnostic

The $\alpha = 0$ diagnostic was analysed separately from the 16-cell core experiment. It uses the coordinate-labelled construction with $\alpha = 0$, so that the deliberately structured coefficient component receives zero weight and the node-specific deviations are not deliberately aligned with the supplied graph.

Contrary to a strict “no systematic benefit” expectation, network-weighted smoothing had lower node-averaged ISE than nodewise smoothing in every replication (Table 5). The relative MISE reductions were 13.29% for the queen graph and 13.41% for the rook graph. The mean paired improvement was 0.004486 for the queen graph (paired MCSE 0.000142) and 0.004527 for the rook graph (paired MCSE 0.000141). Thus, in both cases the mean improvement exceeded twice its paired MCSE by a wide margin.

Table 5: Results of the separate $\alpha = 0$ diagnostic. MISE ratios below one favour network-weighted smoothing.

Graph	MISE ratio	Reduction (%)	Win rate
Queen	0.867	13.29	100%
Rook	0.866	13.41	100%

The residual advantage is consistent with generic partial pooling in addition to any graph-specific borrowing. When the deviations are not deliberately aligned with the supplied graph, the neighbours of a node form an effectively random subset. Shrinking a node’s coefficient vector towards its neighbourhood average can therefore also act as shrinkage towards a common mean. The shared functional intercept and the jointly estimated temporal smoothing parameter may contribute in the same direction. The near-identical results for the rook and queen graphs are consistent with this interpretation, although they do not identify the individual contribution of each mechanism.

This diagnostic sharpens the interpretation of the core results. The comparison with nodewise smoothing combines generic joint shrinkage with any benefit associated with graph-aligned structure. The $\alpha = 0$ reduction of about 13.3% provides a descriptive reference at the high-noise level. The high-noise reductions for the $\alpha = 0.5$ cells are close to this reference. The larger reductions for coordinate-smooth truth with $\alpha = 0.9$ are consistent with a greater benefit when the deviations have stronger coordinate-smooth structure. The smaller reductions for cluster-structured truth with $\alpha = 0.9$ are consistent with the boundary-specific attenuation reported in Section 5.5. These comparisons are descriptive because the $\alpha = 0$ diagnostic uses a different truth distribution and no $\sigma = 0.2$ diagnostic was conducted. They do not by themselves isolate a topology-specific effect of the particular graph edges.

5.6.1 Rewired-graph ablation

To assess whether the advantage of the network-weighted estimator depends on the particular supplied graph edges rather than only on generic joint shrinkage, a fixed degree-preserving rewired-graph control was conducted. The rewired graph preserved the 25 nodes, the rook degree sequence, the total number of edges, and connectivity, while replacing 35 of the 40 rook edges (87.5%). True curves and observations were held fixed within every paired replication;

only the graph supplied to the network-weighted estimator changed.

In both selected high-structure, high-noise settings ($\alpha = 0.9$, $\sigma = 0.5$), the rewired graph increased reconstruction error relative to the correct rook graph. The result was particularly clear for coordinate-smooth truth, where the rewired graph had 9.1% higher aggregate error and was worse in 98.0% of paired replications. For cluster-structured truth, the corresponding increase was 3.3%, and the rewired graph was worse in 81.5% of replications. The paired MCSEs were small relative to the estimated differences in both settings. The MISE ratio in Table 6 is $\text{MISE}_{\text{rewired}}/\text{MISE}_{\text{rook}}$; values above one therefore favour the correct rook graph.

Table 6: Fixed rewired-graph control. Positive differences indicate higher node-averaged integrated squared error (AISE) for the rewired graph than for the correct rook graph.

Truth structure	Mean $\Delta_{\text{rewired-rook}}$	Paired MCSE	MISE ratio	Rewired worse (%)
Coordinate-smooth	0.002404	0.000077	1.091	98.0
Cluster-structured	0.000954	0.000072	1.033	81.5

5.7 Restricted-oracle tuning diagnostic

A restricted-oracle diagnostic was conducted to assess the gap between REML smoothing-parameter selection and the best selection available within a prespecified two-dimensional grid for the graph and time interaction penalties. The functional-intercept smoothing parameter was held fixed at its REML value. The diagnostic used 50 paired replications in four prespecified rook-graph cells. For each simulated dataset, the oracle selected the candidate with the lowest AISE relative to the known true curves. It is therefore not a practically available competing estimator, but a diagnostic benchmark for the smoothing-parameter selection used by the network-weighted estimator.

All 200 REML fits and all 200 restricted-oracle searches completed successfully. In addition, the zero-offset candidate reproduced the REML prediction exactly in every replication, confirming the implementation of the restricted comparison. Table 7 reports the paired REML-minus-oracle AISE differences. Positive values indicate lower AISE for the oracle-selected candidate.

Table 7: Restricted-oracle tuning diagnostic based on 50 paired replications per cell. The reported paired difference is $\text{AISE}_{\text{REML}} - \text{AISE}_{\text{oracle}}$; positive values favour the oracle-selected candidate. Cells with an inadequate search range are reported descriptively only.

Truth	α	σ	Search range	Mean paired difference	Paired MCSE	Oracle lower AISE (%)
Cluster	0.9	0.5	Adequate	0.002661	0.000156	100.0
Coordinate	0.0	0.5	Inadequate	0.001609	0.000191	90.0
Coordinate	0.9	0.2	Inadequate	0.000771	0.000035	100.0
Coordinate	0.9	0.5	Adequate	0.002714	0.000207	100.0

The prespecified coarse-grid boundary audit was passed in the cluster-structured and coordinate-smooth high-noise cells. In these two range-adequate cells, the oracle-selected candidate had lower AISE than the REML fit in every replication. This shows that, within the prespecified

candidate range, REML was not identical to the true-error-optimal choice of the two interaction penalties.

However, the audit classified two cells as range-inadequate. The lower graph-penalty boundary was selected in 26% of coarse searches for the coordinate-smooth low-noise cell and in 22% for the $\alpha = 0$ zero-structured-signal diagnostic, exceeding the prespecified 5% threshold. Thus, the corresponding oracle improvements may be lower bounds, because the best candidate could lie outside the searched range. Following the prespecified decision rule, the grid was not expanded post hoc and the affected cells are not used for formal oracle-comparison conclusions.

Overall, this diagnostic supports a limited conclusion: REML-based tuning can differ materially from the best candidate under true-curve loss, while the magnitude of this gap depends on the candidate range. It does not establish that an oracle procedure is available in practice, nor does it alter the primary network-versus-nodewise comparison.

6 Applications to Observed German Data

The simulation study evaluated reconstruction accuracy under controlled data-generating mechanisms with known true curves. The observed-data applications complement this evidence by demonstrating the implementation on publicly available data in two substantively different graph settings. The DWD temperature data were selected because they provide functional observations at geographically distributed point locations, for which a distance-based nearest-neighbour graph is a natural representation of spatial proximity. The RKI COVID-19 data were selected because they provide county-level functional observations with an administratively meaningful polygon-contiguity graph. Together, the two applications assess whether the workflow can be implemented for both distance-based and areal neighbourhood structures. Because the latent curves are unknown, they are descriptive illustrations rather than external accuracy validations.

6.1 Data and preprocessing

The first application uses daily mean air temperatures from the Climate Data Center of the German Weather Service (DWD) (Deutscher Wetterdienst, 2024). The data were obtained with the `rdwd` R package (Boessenkool, 2026).

For a geographically distributed and computationally manageable illustration, 25 DWD stations with complete daily climate records for 2024 were prespecified and retained as a fixed application design. The analysis uses the daily mean temperature variable `TMK.Lufttemperatur` from 1 January to 31 December 2024. Since 2024 was a leap year, each station contributes 366 observations. The selected stations had the complete common date grid, no duplicate dates, and no missing values. Thus, the data form a 25×366 matrix of observed temperatures, with one curve per station.

6.2 Graph construction and fitted models

Station coordinates were transformed to ETRS89 / LAEA Europe (EPSG:3035) before distances were computed. Each station was connected to its four nearest neighbours, and the resulting

directed nearest-neighbour relation was symmetrised. The final binary graph has 25 nodes, 62 undirected edges, and one connected component. Node degrees range from 4 to 7; edge lengths range from 61.5 to 253.2 km, with a median of 145.2 km. Figure 3 shows the final graph. The six highlighted stations are used for the curve displays below. They were selected to provide geographical coverage of southern, central, and northern Germany rather than on the basis of the fitted results.

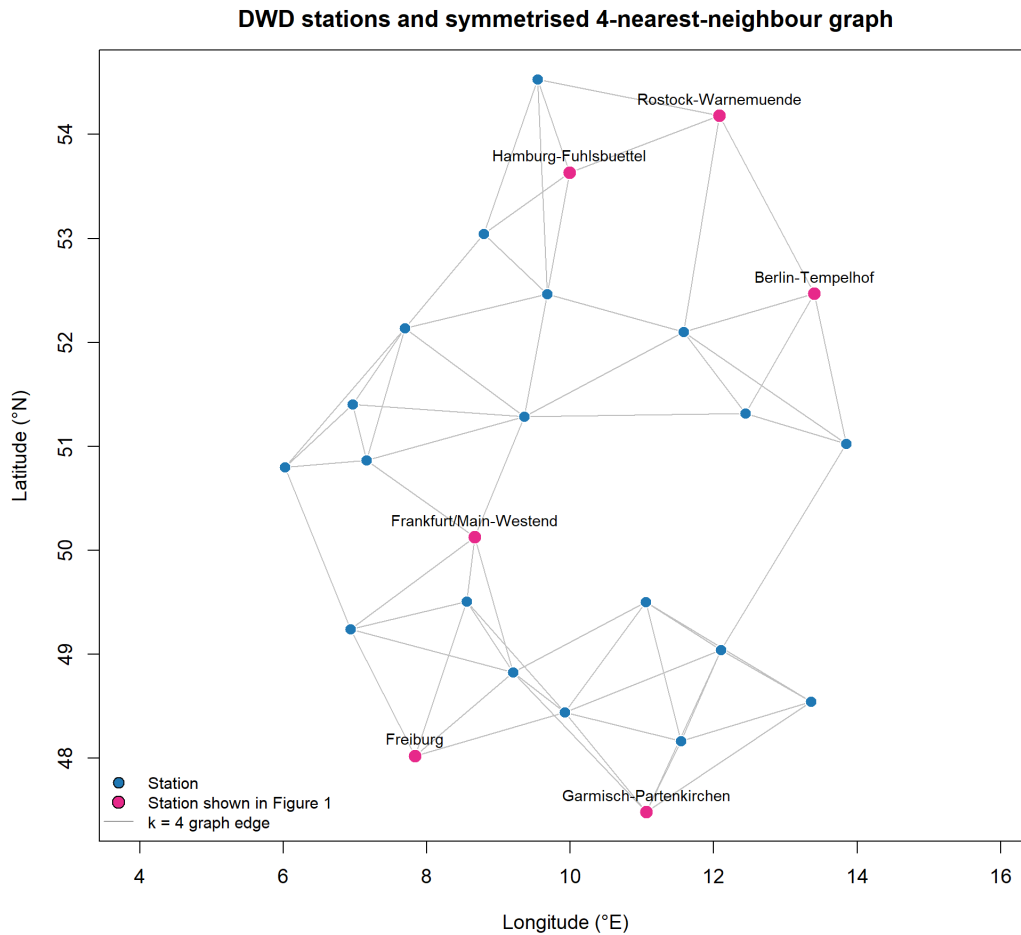


Figure 3: Twenty-five DWD temperature stations and the symmetrised four-nearest-neighbour graph. Pink stations are shown in Figure 4. The graph was constructed from projected coordinates; longitude and latitude are used here only for display.

The dates were linearly rescaled to $[0, 1]$. The nodewise comparator estimates each station curve separately and does not borrow information across stations. The network model estimates the station-specific curves jointly and uses the graph penalty described in Section 3 to discourage differences between the temporal deviations of neighbouring stations. The node-wise comparator fits a separate P-spline with basis dimension $k = 15$ at each station. The network model uses the same graph and the functional additive mixed-model implementation described in Section 3, with a shared temporal intercept basis dimension of 10 and a node-specific temporal deviation basis dimension of 15. Smoothing parameters were selected by REML. These basis dimensions match the final settings used in the simulation study.

6.3 Descriptive comparison of fitted curves

Figure 4 compares the observed temperature series with the nodewise and network-weighted fitted curves for six stations spanning southern, central, and northern Germany. Both estimators capture the broad annual temperature pattern evident in the observed data. The fitted curves are similar but not identical. The network-weighted curves generally preserve the timing of the seasonal peaks and troughs while exhibiting local differences from the nodewise fits, particularly around short-term temperature fluctuations. No selected station shows a qualitatively different annual pattern under network smoothing. Across all 25 stations, the root-mean-square difference between the network and nodewise estimates has a median of 0.542°C and a mean of 0.539°C . The differences range from 0.389°C to 0.786°C .

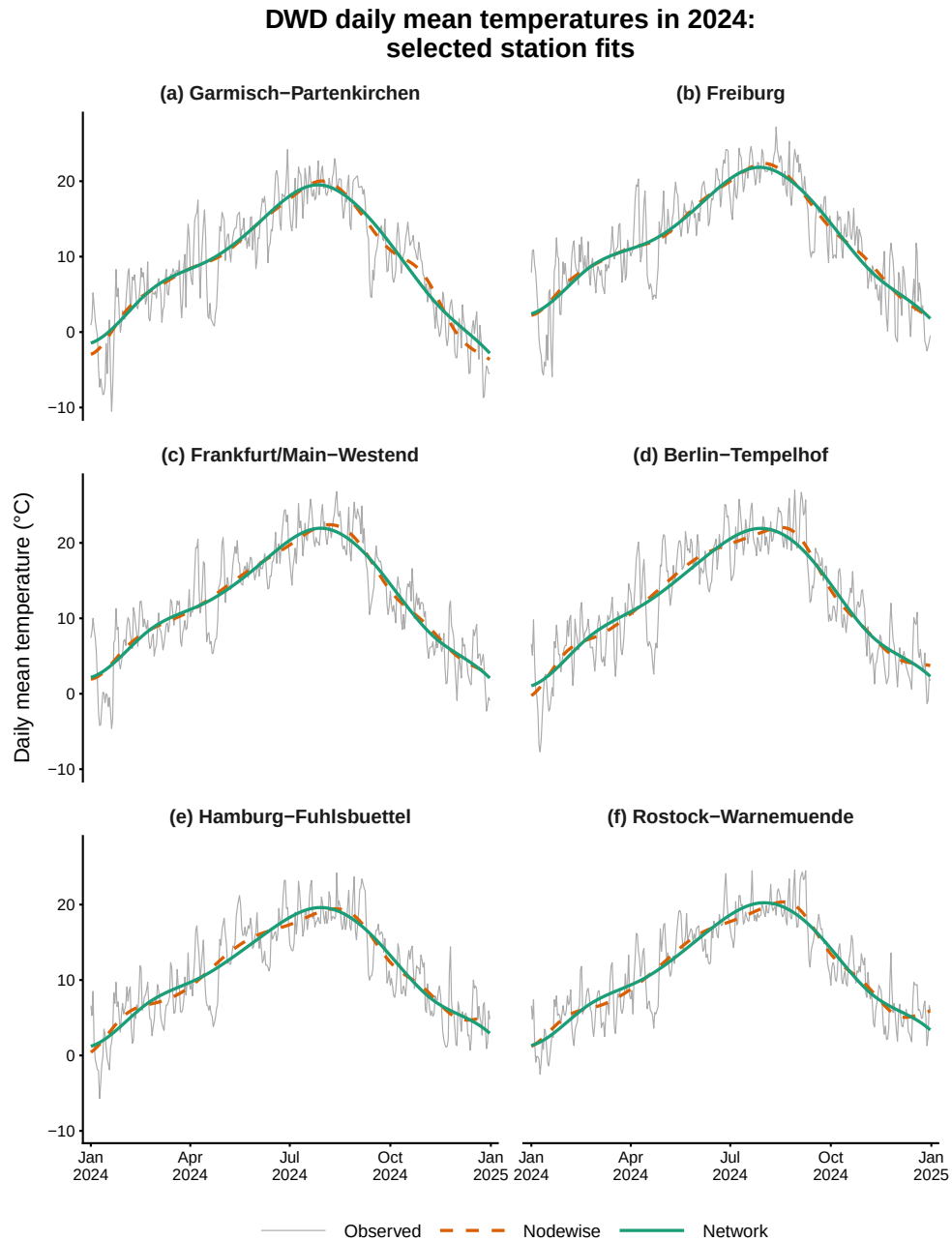


Figure 4: Observed daily mean temperatures in 2024 (grey), separate nodewise fitted curves (orange dashed), and network-weighted fitted curves (green solid) for six selected DWD stations. All panels use the same temperature scale.

Table 8 reports root-mean-square differences between pairs of observed and fitted curves. This is a descriptive regularisation diagnostic rather than an accuracy measure. The latent temperature functions are unavailable, and lower pairwise differences are an intended consequence of smoothing.

For the observed temperatures, station pairs joined by a graph edge are more similar on average than station pairs without an edge. Relative to nodewise smoothing, the network fit reduces the mean edge-pair RMS difference from 1.021°C to 0.845°C . This corresponds to a reduction of 17.3%. The mean non-edge RMS difference decreases from 1.539°C to 1.264°C , corresponding to a reduction of 17.9%. Consequently, the edge-to-non-edge ratio changes only slightly, from 0.664 for the nodewise fit to 0.669 for the network fit.

Table 8: Descriptive pairwise curve-difference diagnostic for the DWD application. RMS differences are measured in $^{\circ}\text{C}$.

Method	Mean edge RMS	Median edge RMS	Mean non-edge RMS	Median non-edge RMS	Edge/non-edge ratio
Observed	1.601	1.500	2.607	2.543	0.614
Nodewise	1.021	0.787	1.539	1.386	0.664
Network	0.845	0.655	1.264	1.105	0.669

The DWD analysis completed with finite fitted curves and shows additional homogenisation of the temperature curves under network regularisation. However, the relative reduction in pairwise differences is almost the same for graph neighbours and non-neighbours. The descriptive diagnostic therefore does not provide evidence that the observed reduction is specific to the chosen graph topology. Together with the $\alpha = 0$ diagnostic in Section 5.6, this supports a cautious interpretation of the real-data behaviour as combining graph-structured borrowing with more general joint shrinkage. Establishing reconstruction accuracy or a topology-specific effect would require a separate validation design with a known or independently evaluated target.

6.4 RKI COVID-19 data and preprocessing

A second application considers weekly reported COVID-19 incidence at the German county level. Case data were obtained from the public repository of the Robert Koch Institute (RKI) (Robert Koch Institute, 2026). The source repository is continuously updated; therefore, the downloaded raw CSV file and an input manifest were retained as part of the reproducibility materials. This application demonstrates the workflow with an areal-contiguity graph. Since the underlying incidence functions are not observed, it is not an external accuracy validation.

The analysis covers the 157 calendar weeks from 2 March 2020 to 5 March 2023, where weeks run from Monday to Sunday. Records were aggregated by the official county identifier (IdLandkreis) and by week of Meldedatum. The reported case counts were summed as supplied by the RKI, including retrospective corrections; county-week combinations absent from the source were assigned zero reported cases. The twelve Berlin boroughs were aggregated to the Berlin county-level identifier before matching the case data to the polygon layer. This yielded a complete panel of 400 German administrative units and $400 \times 157 = 62,800$ county-week observations. The resulting weekly count totals contained no negative values, and 1,826 county-week observations had zero reported cases.

County polygons and population denominators were obtained from the BKG VG250-EW administrative-boundary archive with population figures as of 31 December 2020 (Bundesamt für Kartographie

und Geodäsie, 2020). After excluding unpopulated administrative areas in the BKG layer, the 400 populated polygon identifiers matched the county identifiers in the case panel one-to-one. Weekly reported incidence was calculated as reported cases per 100,000 residents using the 2020 population denominator. No negative incidence values occurred. The fitted response was

$$\log\{1 + \text{weekly reported cases per 100,000 residents}\}.$$

The transformation is defined for zero case counts and reduces the influence of very large weekly incidences. All fitted values in this application are therefore interpreted on the log-incidence scale.

6.5 RKI graph construction and fitted models

A binary Queen-contiguity graph was constructed from the matched county polygons. Thus, two counties were treated as neighbours when their polygons shared at least one boundary point, including a vertex. The resulting nationwide graph contains 400 nodes, 1,044 undirected edges, no isolated nodes, and one connected component.

An attempt to fit the network model to all 400 counties exceeded the available memory in the underlying `mgcv` fitting routine. The full German graph was nevertheless constructed and audited before model fitting. To obtain a computationally feasible and reproducible polygon-based illustration, the analysis was restricted to the induced subgraph of Bavaria, defined by county identifiers beginning with 09. This restriction was made for computational feasibility rather than on the basis of fitted results. The Bavarian subgraph contains 96 counties, is connected, and has no isolated nodes. The response matrix used for fitting consequently has dimension 96×157 .

The nodewise comparator fits a separate P-spline with basis dimension $k = 15$ to each county. The network model uses the Bavarian Queen-contiguity graph, a shared temporal intercept basis dimension of 10, and node-specific temporal deviation basis dimension of 15. The P-spline specifications correspond to `bs = "ps"` and `m = c(2, 1)` in the implementation. Smoothing parameters were selected by REML. These basis dimensions match the final settings of the simulation study.

6.6 Descriptive comparison of RKI fitted curves

All 96 nodewise fits converged, and the network model also converged. The network model had an effective degrees of freedom of 445.2. Across all Bavarian counties and weeks, the mean absolute difference between the network and nodewise fitted values was 0.224 on the log-incidence scale. The largest absolute difference among fitted county-week values was 2.268. These quantities describe disagreement between two estimators; they are not measures of estimation error because a true incidence curve is unavailable.

Figure 5 maps, for each county, the mean absolute difference between the network and node-wise fitted curves across the 157 weeks. Higher values indicate a larger discrepancy between the two fitted curves. The magnitude of this disagreement is spatially heterogeneous across Bavaria: some counties show comparatively small differences between the estimators, whereas others have mean absolute differences near the upper end of the displayed range. The map must not be interpreted as a map of incidence, reconstruction error, or a causal effect of network

smoothing. Differences may arise from both graph-structured borrowing and the general joint regularisation of the network model.

Difference between network and nodewise fitted curves

Mean absolute difference across the 157 weekly observations

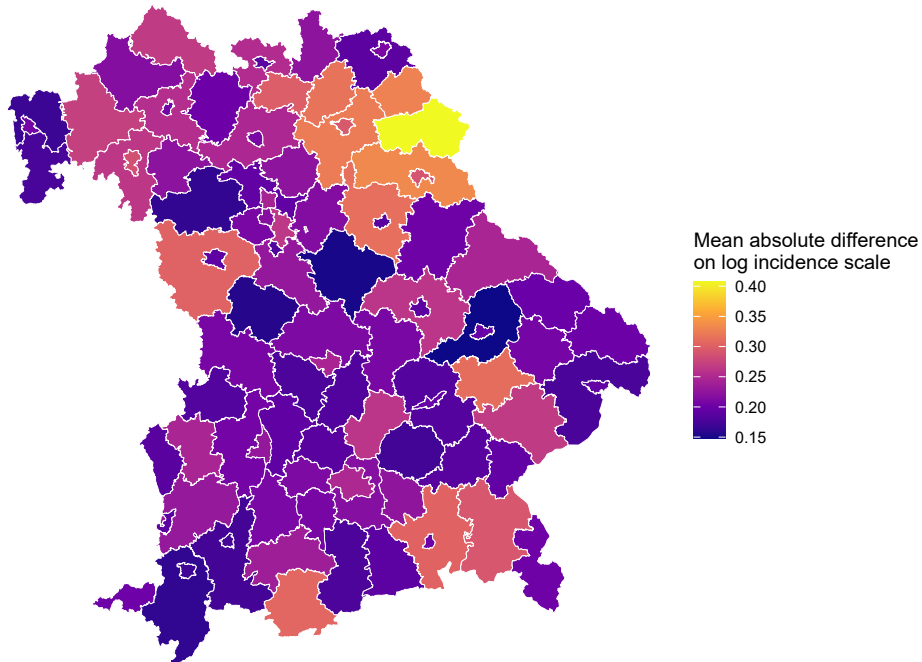


Figure 5: Mean absolute difference between the network-weighted and nodewise fitted curves, averaged across 157 weekly observations for Bavarian counties. Values are on the log-incidence scale; higher values indicate greater disagreement between the estimators.

Figure 6 compares observed weekly incidence with the two fitted curves for Berchtesgadener Land, Munich, Nuremberg, and Würzburg. Both estimators retain the broad timing of the observed epidemic waves, while they differ locally in fitted levels and smoothness. The differences between the fitted curves are most visible during periods of rapid change and around pronounced incidence peaks, whereas their broad temporal trajectories remain similar across the four counties. These displays are illustrative only and do not establish that either estimator is closer to an unobserved target function. The four counties were selected to illustrate different locations and urbanisation contexts within Bavaria and were not selected on the basis of the fitted differences.

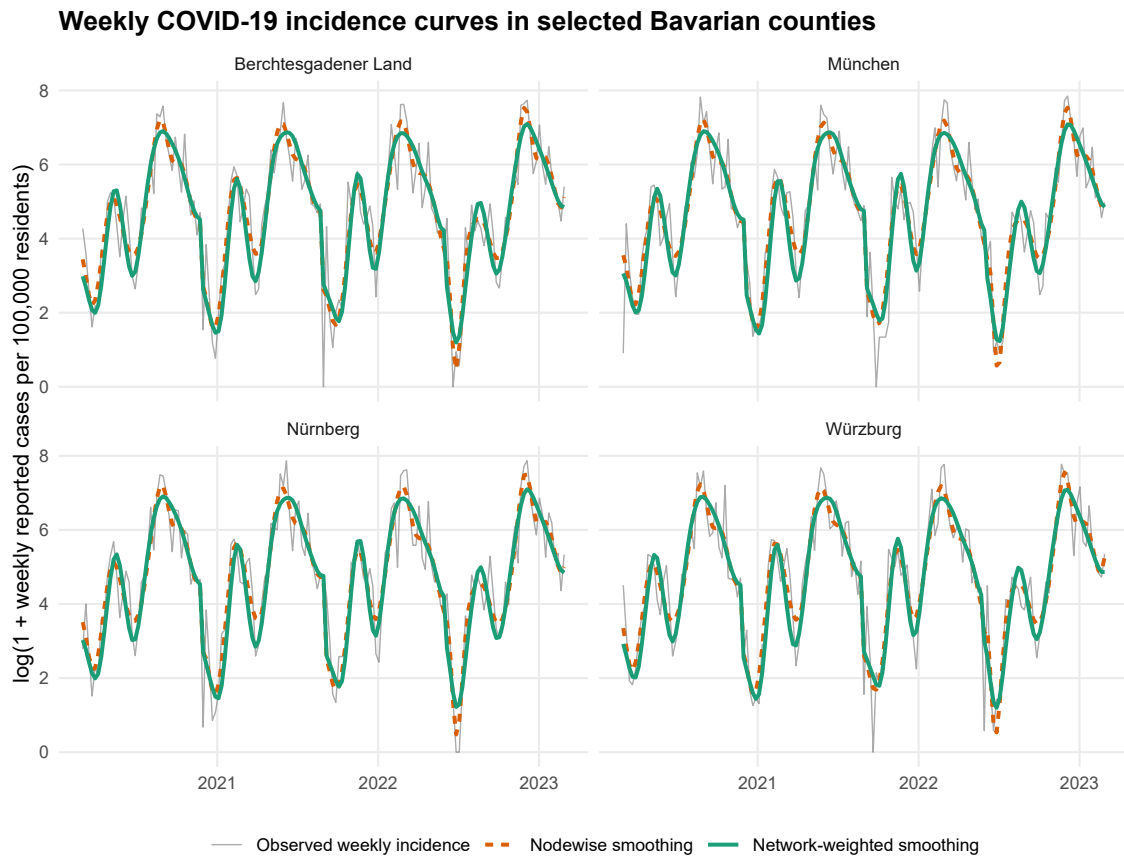


Figure 6: Observed weekly COVID-19 incidence (grey) and fitted curves from separate node-wise smoothing (orange, dashed) and network-weighted smoothing (green, solid) for four illustrative Bavarian counties. The vertical scale is the natural logarithm of one plus weekly reported cases per 100,000 residents.

The RKI application shows that the workflow can also be implemented for areal functional data and a polygon-contiguity graph. It does not provide evidence of a topology-specific accuracy advantage. In particular, the mapped differences only compare the two fitted estimators and cannot distinguish graph-specific borrowing from more general joint shrinkage. Conclusions about reconstruction accuracy are therefore based on the simulation study rather than on this observed-data illustration. The restriction to Bavaria also illustrates a current computational limitation of the implementation for large areal graphs. Scaling the network fit to the full German county graph would require further computational development.

7 Discussion

This thesis considered the reconstruction of node-specific functional data when the observation units are linked by a supplied graph. The proposed joint estimator combines a common functional mean, temporal regularisation of node-specific deviations, and a graph penalty that encourages neighbouring nodes to have similar curves. Its implementation uses functional additive mixed-model infrastructure and permits the supplied graph to be translated directly into the neighbourhood structure required by the fitting procedure.

In the core simulation, the complete joint network-weighted estimator had lower aggregate MISE than separate nodewise smoothing in all 16 design cells. The relative MISE reduction ranged from 1.4% to 20.5%, and the network estimator won in between 72.0% and 100% of the paired replications. The largest improvements occurred in the high-noise coordinate-smooth scenarios with $\alpha = 0.9$. This pattern is compatible with the motivation for joint smoothing: when node-specific curves contain a strong smooth component and the observations are relatively noisy, borrowing information across nodes can reduce estimation variance. Thus, within the examined $N = 25$ settings, the complete joint network-weighted estimator improved reconstruction relative to separate nodewise smoothing.

The magnitude of this improvement depended on the type of underlying signal. For cluster-structured truth, the gains were more modest when $\alpha = 0.9$, particularly at low noise. The cluster-boundary diagnostic provides a complementary explanation. At $\sigma = 0.5$, the boundary difference-in-differences was positive beyond twice its paired MCSE in all four cluster-structured cells. Thus, although the network estimator still had lower average error at boundary nodes than the nodewise estimator, its relative advantage was smaller there than at matched interior nodes. This is consistent with a bias–variance trade-off: graph regularisation can stabilise estimated curves within a cluster, but may attenuate differences across a discontinuity.

The comparison of rook and queen graphs does not support a general conclusion that a denser graph is preferable. Their results were often close, and their relative ordering varied across cells. In particular, the queen graph was not uniformly superior in the cluster-structured scenarios. This suggests that the usefulness of a supplied graph depends on how its edges relate to the underlying functional signal, rather than on edge density alone.

The separate $\alpha = 0$ diagnostic further qualifies the interpretation of the main comparison. In that setting, the deliberately structured coefficient component received zero weight and the node-specific deviations were not deliberately aligned with the supplied graph. Nevertheless, the network estimator achieved an aggregate MISE reduction of approximately 13.3% relative to nodewise smoothing for both graph specifications and had lower node-averaged ISE in every replication. This result is consistent with generic joint shrinkage contributing to the compari-

son with nodewise smoothing, in addition to any gain associated with graph-aligned borrowing. Consequently, the core simulation establishes an advantage of the complete joint estimator under the examined conditions, but the nodewise comparison alone cannot separate generic joint shrinkage from the benefit of alignment with the particular supplied graph edges.

The rewired-graph ablation provides complementary evidence on the role of the particular supplied edges. Within each paired replication, the true curves and observations were held fixed while the correct rook graph was replaced by a fixed degree-preserving rewired graph. In both selected high-structure, high-noise settings, the rewired graph increased reconstruction error: by 9.1% for coordinate-smooth truth and by 3.3% for cluster-structured truth. Thus, although the $\alpha = 0$ diagnostic is consistent with a contribution from generic joint shrinkage, these selected comparisons provide evidence that alignment between the supplied graph and the underlying signal can provide an additional reconstruction benefit.

The two applications illustrate complementary aspects of the workflow. For the DWD temperature data, both estimators recovered the annual temperature pattern, while the network fit produced slightly smaller fitted differences between stations. However, edge-pair and non-edge-pair differences decreased by almost the same proportion. The DWD analysis therefore provides no evidence that this reduction was specific to the chosen graph topology. For the RKI incidence data, the analysis demonstrates that the method can be fitted with an areal-contiguity graph and a transformed epidemiological response. Because the latent curves are unavailable in both applications, these analyses are descriptive illustrations rather than external accuracy validations. The restriction of the RKI analysis to Bavaria also makes the current computational limitation of the implementation explicit.

Several limitations define the scope of these conclusions and the settings to which they can be transferred. The simulation uses a small regular lattice, Gaussian observations, a common dense time grid, and binary undirected graphs. It represents each node by one noisy curve. Under the assumed independent Gaussian errors on a common grid, averaging multiple replicate curves at a node is equivalent to reducing the noise standard deviation; this equivalence need not hold under correlated errors or irregular observation schemes. The estimator is evaluated for reconstruction at observed nodes and does not address prediction at unobserved locations.

The prespecified $N = 100$ timing gate showed that individual fits were numerically successful in two representative cells, but that even the smallest planned performance comparison, with $B_{100} = 25$ attempted replications per cell, exceeded the eight-hour serial-runtime budget. Thus, this thesis makes no reconstruction-accuracy claim at $N = 100$. The restricted-oracle diagnostic is likewise limited to a fixed candidate range. Two of its four cells were classified as range-inadequate because the coarse search frequently selected the lower graph-penalty boundary. Those cells are not used for formal oracle-comparison conclusions.

Finally, the $\alpha = 0$ diagnostic was conducted only at the high-noise setting. The rewired-graph ablation provides direct evidence on graph-edge alignment, but it uses one fixed degree-preserving rewired graph and two selected scenarios; it therefore does not characterise all forms of graph misspecification.

8 Conclusion

This thesis developed and evaluated a joint network-weighted smoothing approach for functional data observed at graph-connected nodes. The method combines a common functional mean, temporal regularisation of node-specific deviations, and a graph-based penalty, and can be implemented in a reproducible functional additive mixed-model workflow.

Across the 16 core $N = 25$ simulated settings, the complete joint estimator improved reconstruction relative to independent nodewise smoothing. Relative MISE reductions ranged from 1.4% to 20.5%, with the largest improvements occurring in high-noise coordinate-smooth settings with $\alpha = 0.9$. This is consistent with the motivation for joint smoothing: when node-specific curves share smooth structure and observations are noisy, borrowing information across nodes can reduce reconstruction error.

The separate $\alpha = 0$ diagnostic qualifies the interpretation of this main comparison. In that setting, the coordinate-smooth component received zero weight and the node-specific deviations were not deliberately aligned with the supplied graph. Nevertheless, the network estimator achieved an aggregate MISE reduction of approximately 13.3% relative to nodewise smoothing for both graph specifications and had lower node-averaged ISE in every replication. Thus, this comparison is consistent with generic joint shrinkage contributing to the observed advantage in addition to any gain associated with graph-aligned borrowing.

The fixed rewired-graph control provides complementary evidence on the role of the particular supplied edges. Holding the true curves, observation errors, node degrees, total edge count, and graph connectivity fixed, replacing 87.5% of the rook edges increased aggregate MISE in both selected high-structure, high-noise settings. Relative to the correct rook graph, aggregate MISE increased by 9.1% for coordinate-smooth truth and by 3.3% for cluster-structured truth; the rewired graph was worse in 98.0% and 81.5% of paired replications, respectively. Hence, in these examined settings, the advantage of the complete joint estimator is not explained solely by generic joint shrinkage: alignment between the supplied graph edges and the functional signal can provide an additional reconstruction benefit. This result is limited to one fixed rewired graph and two selected scenarios and does not establish a universal effect for all forms of graph misspecification.

In both empirical applications, the specified models fitted successfully and produced finite predictions; however, they do not provide external accuracy validation. Overall, the results support the complete joint approach as a useful reconstruction method when node-specific curves are expected to share structure and a scientifically motivated graph is available. Its use should be accompanied by sensitivity analyses and a cautious interpretation of any apparent advantage as topology-specific. These conclusions concern reconstruction at observed nodes on the small regular graph settings examined here.

A Appendix

A.1 Supplementary implementation details

The function `graph_to_nb()` accepts an adjacency matrix, an `igraph` object, or polygonal `sf` data and returns the named neighbourhood-list representation required by the MRF basis. A numeric matrix input must be square, symmetric, and free of missing values. Diagonal entries are ignored, and each non-zero off-diagonal entry is treated as an edge; edge magnitudes are therefore not retained.

For an `sf` input, the user must provide one unique polygon identifier per feature, either as a column name or as a vector. Polygon contiguity is constructed using `spdep::poly2nb()` with queen contiguity by default. The sequential neighbour indices returned by `poly2nb()` are translated into the user-supplied polygon identifiers, which become the names of the neighbourhood list.

The fitting interface accepts both grid-based `tf::tfd` objects and basis-represented `tf::tfb` objects. A `tf::tfb` input is first converted to `tf::tfd` while retaining its curve identifiers. The curves must then be defined on a common model grid. Curve identifiers must be unique and agree exactly with the graph-node names before the curves are reordered to graph order and passed to the common fitting backend.

The R package, simulation drivers, application scripts, and released result artifacts are available at <https://github.com/Siyiliu41/network-weighted-functional-smoothing>.

A.2 Absolute MISE values for descriptive reference methods

Table 9 reports mean integrated squared error (MISE) over 200 replications for all fitted procedures. The raw, pooled, and nodewise procedures do not use the supplied graph and are therefore reported once for each truth–signal–noise combination. The network-weighted estimator is reported separately under the rook and queen graph specifications. The raw and pooled procedures are descriptive reference methods; the primary paired comparison in the main text is between nodewise and network-weighted smoothing.

Table 9: Mean MISE for the raw, pooled, nodewise, and network-weighted procedures. Network-weighted results are shown separately for the supplied rook and queen graphs.

Truth	α	σ	Raw	Pooled	Nodewise	Network (rook)	Network (queen)
Cluster	0.5	0.2	0.04008	0.36934	0.00702	0.00671	0.00670
Cluster	0.5	0.5	0.25051	0.37053	0.03427	0.02991	0.02996
Cluster	0.9	0.2	0.04003	0.40014	0.00693	0.00683	0.00681
Cluster	0.9	0.5	0.25020	0.40130	0.03356	0.02935	0.03002
Coordinate	0.5	0.2	0.03982	0.33083	0.00694	0.00658	0.00659
Coordinate	0.5	0.5	0.24888	0.33207	0.03388	0.02936	0.02936
Coordinate	0.9	0.2	0.03995	0.33118	0.00690	0.00659	0.00660
Coordinate	0.9	0.5	0.24971	0.33236	0.03312	0.02635	0.02691

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AI Usage Documentation (KI-Hilfsmittel)

I confirm that I take full responsibility for the content of this thesis, the submitted code, and all reported methodological decisions, analyses, results, and conclusions.

Generative artificial intelligence tools, specifically ChatGPT by OpenAI, were used as supporting tools during the preparation of this thesis. Their use included explanations of statistical and programming concepts, assistance with the structure and clarity of scientific writing, language editing, suggestions for code documentation and debugging, and orientation during literature exploration. Such support was used in connection with the Introduction, Theoretical Background, Methods, Simulation Study, Results, Applications, Discussion, Conclusion, and reproducibility materials.

AI-generated suggestions for text and code were not adopted without review. I checked, revised, and adapted them to the actual model specification, the implemented R code, the analyses carried out, and the cited primary sources. In particular, I verified the mathematical formulation of the estimator, the simulation design, the implementation, the quality-control procedures, and the interpretation of the results.

Generative AI was not used to generate numerical results or as a substitute for statistical analysis. All simulations, empirical applications, tables, and figures reported in this thesis were produced using R scripts that I executed and checked. The scientific claims and conclusions of this thesis are based on these reproducible analyses and on the cited literature.

Declaration of Authorship

I hereby declare that I prepared this thesis independently and used no sources or aids other than those explicitly indicated in the thesis. All direct and indirect sources, including the use of generative AI tools, have been appropriately disclosed.

Munich, 31 August 2026

Siyi Liu